Table 1: Run 3 analysis selection in application order.

| Order | Requirement | Selection | Stage |
|---|---|---|---|
| 1 | Golden JSON | Data only: require the run and luminosity section to be included in the era-specific Golden JSON certification mask. | $*$ |
| 2 | Orthogonal MC era split | MC only in 2024–2026: assign each event deterministically to exactly one era using run, luminosity block, event and dataset identifiers (seed 12345). The fractions follow the target luminosities 110:111:26 for 2024:2025:2026, preventing event reuse across eras. | $*$ |
| 3 | Generator VBF phase space | MC only in 2024–2026, for DY $105 < m_{\mu\mu} < 160$ samples. Remove hard-process leptons within $\Delta R < 0.3$ of GenJets and define GenVBFFilter from the two leading remaining GenJets: pass when $m_{jj} > 300$ GeV. The inclusive samples DYto2Mu__MLL105To160 and DYto2Mu__MLL__105to160__amcatnloFXFX use GenVBFFilter= 0; their FlashSim aliases follow the same rule. DYto2Mu__MLL105To160__VBFFiltered and DYto2Mu__MLL__105to160__amcatnloFXFX__Fil__VBF use GenVBFFilter= 1. The flag is stored at skim level and the complementary cuts are applied during histogram routing. | $*$ / † |
| 4 | Event quality | Logical AND of the recommended era-dependent NanoAOD MET filters. | $*$ |
| 5 | Muon momentum | For each muon use the beam-spot-constrained (BSC) $p_T$ when its BSC fit $\chi^2 < 30$; otherwise use the standard NanoAOD $p_T$. ScaRe scale corrections are applied to data and MC, resolution smearing to MC, and FSR recovery is included before selections and invariant-mass construction. | $*$ |
| 6 | Single-muon trigger | Require HLT__IsoMu24 online and at least one selected offline muon with corrected $p_T > 26$ GeV matched to a trigger object with ID 13, filter bit 8 and $\Delta R < 0.4$. | $*$ / † |
| 7 | Muon skim selection | Exactly two ScaRe- and FSR-corrected muons with the Nano/BSC momentum choice above, $p_T > 15.0$ GeV, $\lvert\eta\rvert < 2.4$, medium ID and pfIsoId $\geq 2$. | $*$ |
| 8 | Dimuon skim window | $50 < m_{\mu\mu} < 200$ GeV, evaluated with the selected corrected muon momenta. | $*$ |
| 9 | Electron veto | No electron with $p_T > 20$ GeV, $\lvert\eta\rvert < 2.5$ and MVA Iso WP90. | $*$ |
| 10 | Muon analysis selection | Opposite-sign muons; both with corrected $p_T > 20$ GeV, medium ID and pfIsoId $\geq 2$. The online HLT__IsoMu24 decision and the requirement that at least one of the two offline muons is trigger-matched are retained. | † |
| 11 | Low-pT muon quality | A selected muon with corrected $p_T < 30$ GeV must additionally satisfy tight ID and pfIsoId $\geq 4$ when the corresponding category is used. | † |
| 12 | Jet selection | Jets with $p_T > 20.0$ GeV after jet-energy scale and resolution corrections, $\lvert\eta\rvert < 4.7$, tight jet ID, outside the era-dependent detector horn/veto region and $\Delta R(j,\mu) > 0.4$ from both selected muons. | $*$ |
| 13 | b-jet veto | Within $\lvert\eta\rvert < 2.5$, require zero medium-tagged jets and at most one loose-tagged jet. Working points are era dependent; the tagger is ParticleNet (PNet) for 2022–2023BPix and UParTAK4 for 2024–2026. | † |
| 14 | Mass regions | Inclusive: $60 < m_{\mu\mu} < 150$ GeV; Z sideband: 70–110 GeV; signal fit: 115–135 GeV; H sidebands: 110–115 or 135–150 GeV. | † |
| 15 | Baseline category | Full event selection (Golden JSON for data or MC-specific filters, MET quality and electron veto) plus the two-muon analysis selection, HLT__IsoMu24, at least one offline muon matched to the fired online trigger, and the b-jet veto. | † |
| 16 | VBF topology and category | At skim level, among selected jets outside the horn, define the pair with the largest $m_{jj}$ satisfying $m_{jj} \geq 400$ GeV and $\lvert\Delta\eta_{jj}\rvert \geq 2.5$. At histogram level, the VBF category requires baseline and leading/subleading VBF-jet $p_T \geq 35/25$ GeV. | $*_{\text{def.}}$ / † |
| 17 | ggF categories | Baseline events failing the VBF definition, split into 0-jet, 1-jet and $\geq$ 2-jet categories when requested. | † |

**Stage notation:** $*$ applied during NanoAOD skimming; † applied when producing analysis histograms. Requirements marked with both symbols are enforced at both stages; $*_{\text{def.}}$ means that the object or flag is defined but not used to reject events at skim level. Object calibrations and systematic variations are propagated before the corresponding selections.

Table 2: Reco-jet/GenJet component split for DY, EWK, ggF and VBF MC samples. This classification is orthogonal to the analysis category.

| Reco-jet multiplicity | Component | Definition |
|---|---|---|
| Any | Matching rule | A selected reconstructed jet is hard-scatter matched when its NanoAOD GenJet index is non-negative. A negative index means that no GenJet is matched and the reconstructed jet is classified as a pileup (PU) jet. |
| 0 jets | 0J Hard | No selected reconstructed jet; the event forms the exclusive zero-jet component. |
| 1 jet | 1J Hard / 1J PU | Classify the leading selected reconstructed jet as hard or PU using its GenJet match. |
| $\geq$ 2 jets | 2J Hard / 2J PU1 / 2J PU2 | Count unmatched jets among the first two selected reconstructed jets: respectively 0, 1 or 2 PU jets. |
| VBF pair | VBF Hard / VBF PU1 / VBF PU2 | Apply the same 0/1/2 unmatched-jet count specifically to the two jets forming the selected VBF pair. |

The generic 0/1/2 matching flags are process independent. Component outputs are produced for the configured DY/EWK/ggF/VBF process families when the jet–GenJet splitting workflow is enabled.